

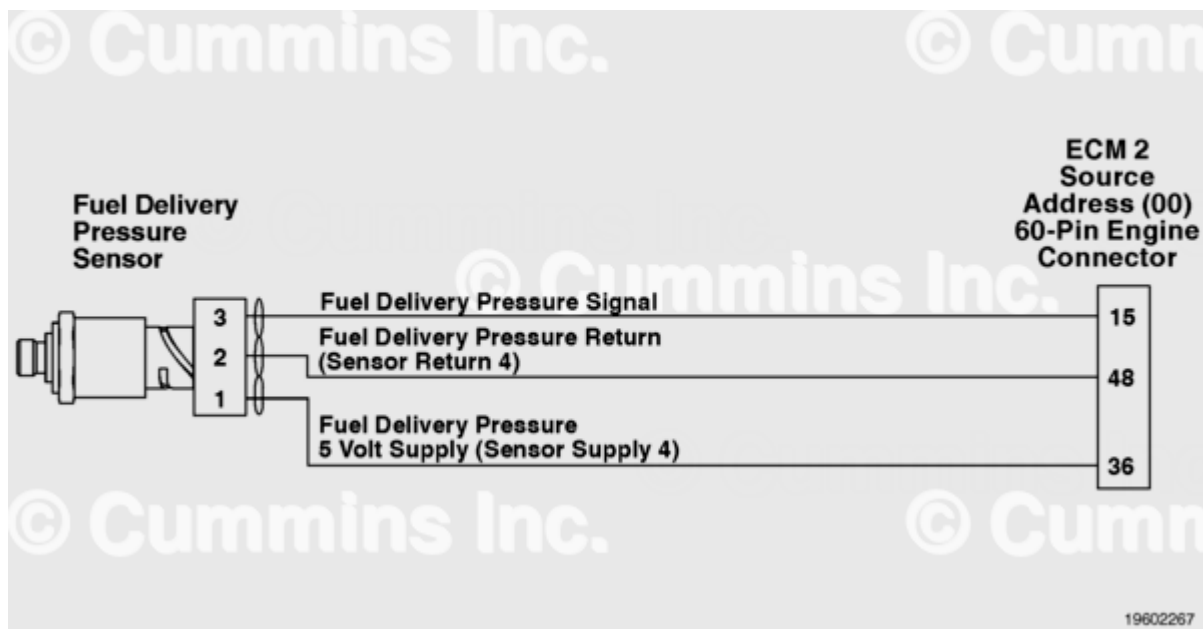
Fault Code: 4615

Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Most Severe Level

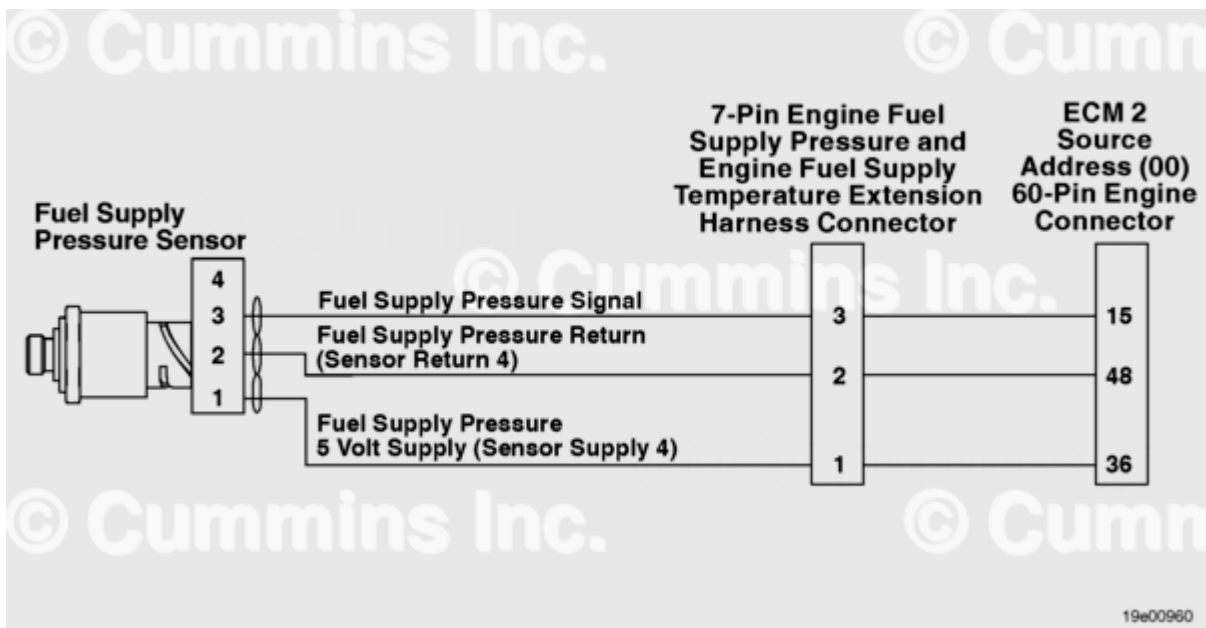
 Printable Version

Overview

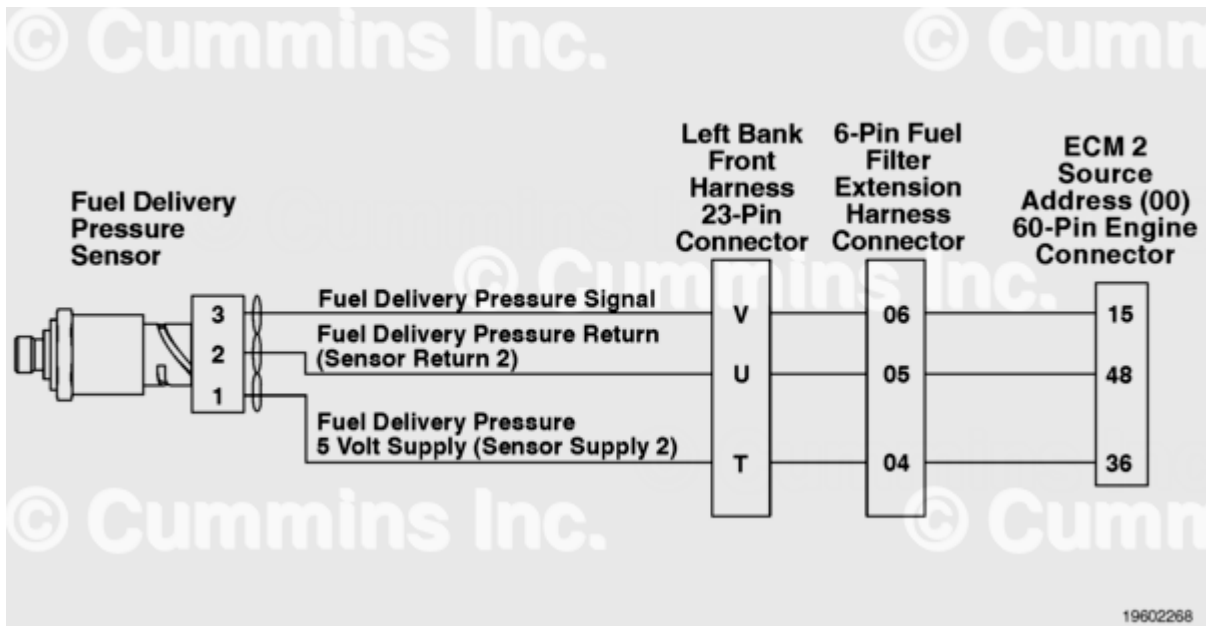
Codes	Reason	Effect
Fault Code: 4615 PID(P): SPN: 94 FMI: 0/0 Lamp: Amber SRT:	Fuel Pump Delivery Pressure - Data Valid But Above Normal Operating Range - Most Severe Level. Fuel pump supply pressure very high.	Possible engine speed derate. Low power or engine smoke.



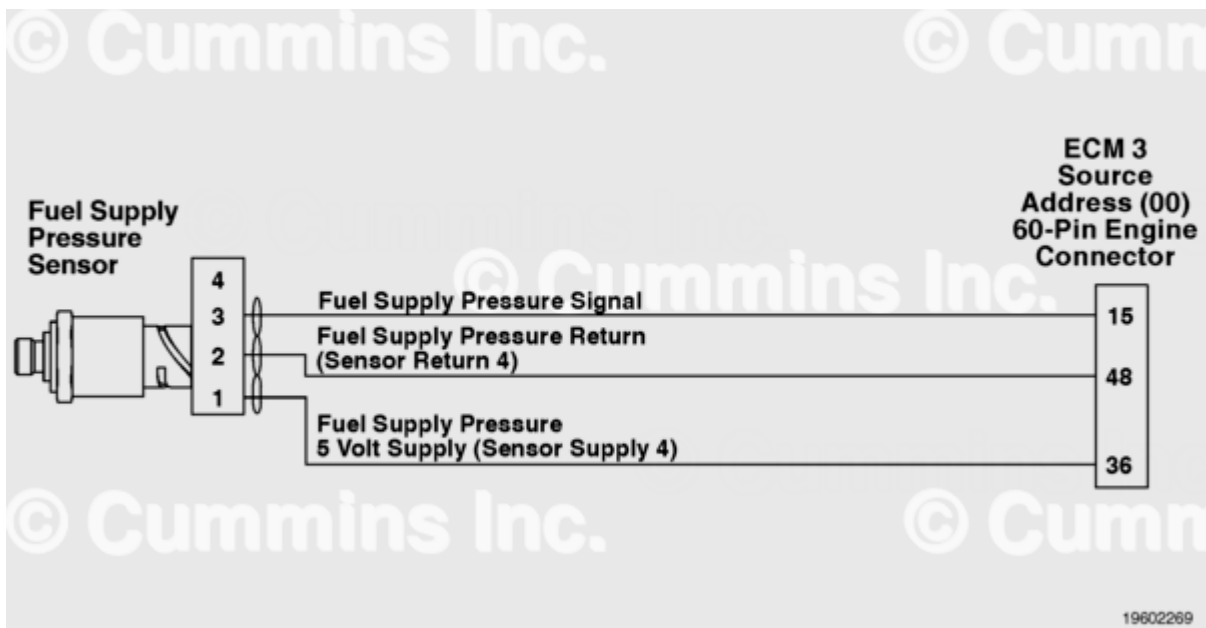
QSK38 CM2150 Industrial - Fuel Delivery Pressure Sensor Circuit



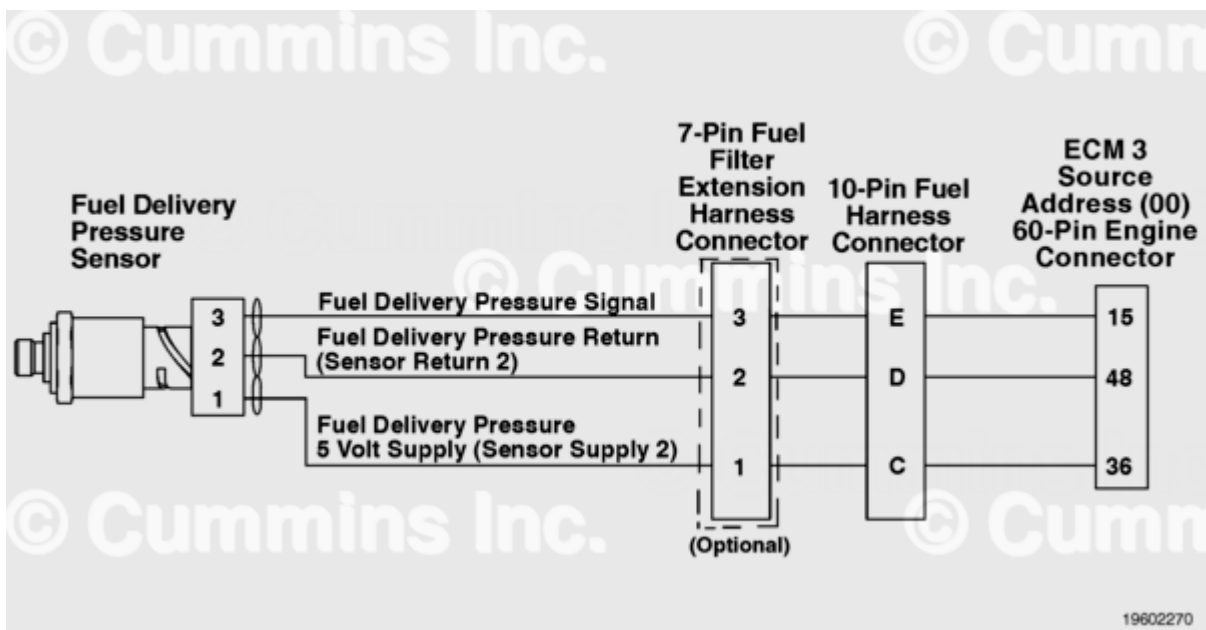
QSK38 CM2150 Power Generation/QSK38 Power Generation (Military Application) - Fuel Delivery Pressure Sensor Circuit



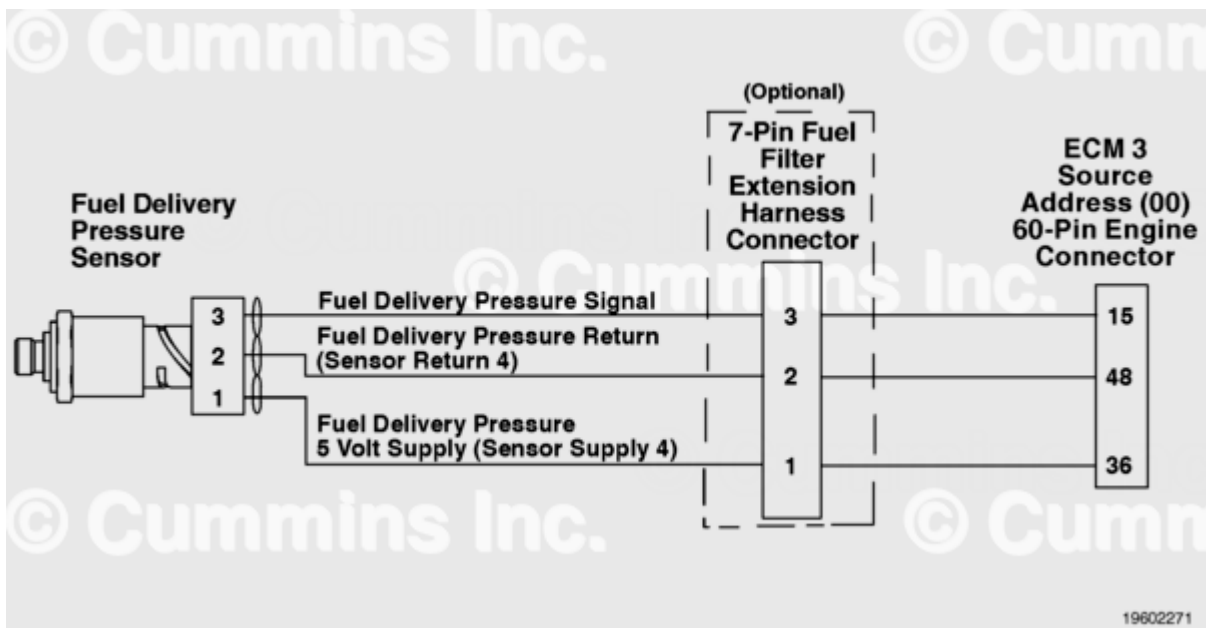
QSK38 CM2150 Marine - Fuel Delivery Pressure Sensor Circuit



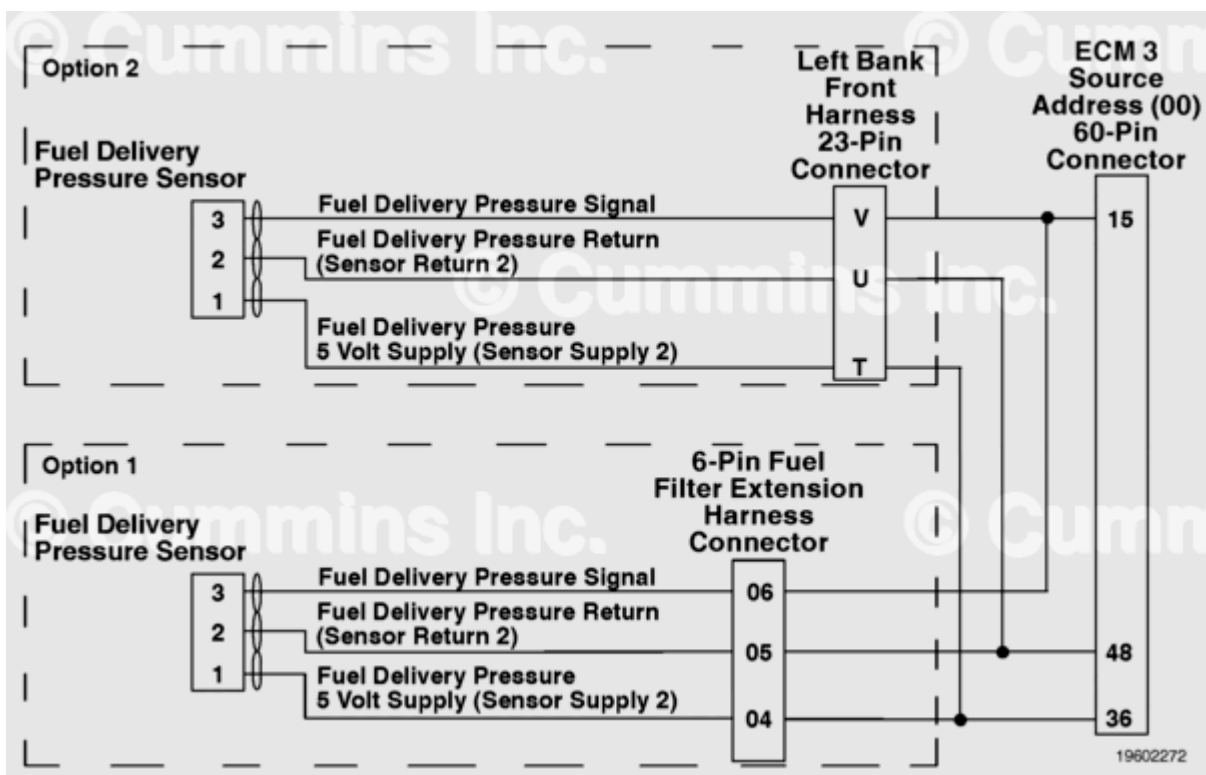
QSK50 CM2150 Power Generation without Advanced Engine Monitoring/QSK50 CM2150 Power Generation with Advanced Engine Monitoring and QSK60 CM2150 Power Generation - Fuel Delivery Pressure Sensor Circuit



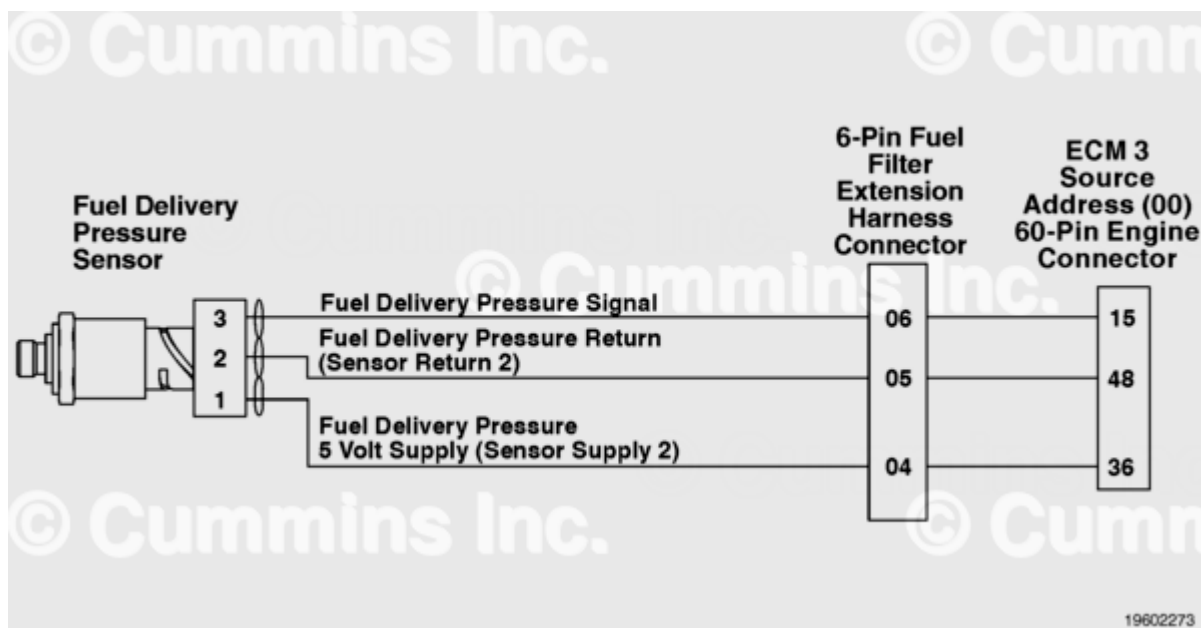
QSK50 CM2150 Industrial - Fuel Delivery Pressure Sensor Circuit



QSK60 CM2150 Industrial - Fuel Delivery Pressure Sensor Circuit



QSK50 CM2150 Marine - Fuel Delivery Pressure Sensor Circuit



QSK60 CM2150 Marine/QSK60 CM2150 Drill Rig - Fuel Delivery Pressure Sensor Circuit

Circuit Description

The fuel delivery pressure sensor is used by the engine control module (ECM) to monitor fuel delivery pressure directly before the Stage 2 filter. The ECM monitors the voltage on the fuel delivery pressure SIGNAL pin and converts it to a pressure value.

Component Location

The fuel delivery pressure sensor is located in the Stage 2 fuel filter head.

Conditions For Running The Diagnostics

- The engine speed **must** be above 500 RPM for 10 seconds, and fueling **must** be above 20 mg/stk for 5 seconds before a diagnostic decision begins to be made.
- The fuel temperature **must** be greater than, or equal to -10 °C

Conditions For Setting The Fault Codes

When the fuel supply pressure is greater than 1200 kPa (174 psi) for more than 3 seconds.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE lamp immediately when the diagnostics detects high fuel supply pressure.
- This fault code also activates an engine speed derate (1500 RPM) that attempts to bring down the fuel supply pressure at or below 1000 kPa. The speed derate is progressive and it is possible that the engine speed may pause above 1500 RPM if the derate is successful at bringing fuel supply pressure below 1000 kPa.

Conditions For Clearing The Fault Code

FC4615 is a latching fault code. The diagnostic only clears errors if the fuel supply pressure is below 850 kPa. Once the fault code is invoked, the only way to clear it would be to perform the following steps:

- Turn off engine and follow the troubleshooting steps for resolution.
- Restart engine, and run the engine at rated for a few minutes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The fault code status displayed by INSITE™ electronic service tool will change to INACTIVE immediately after the diagnostic runs and passes.
- The Reset All Faults command in INSITE™ electronic service tool can be used to clear active and inactive faults.

Shoptalk

There are multiple ECMs for the engine models included in this manual. Each ECM has an individual source address that displays when INSITE™ electronic service tool is connected. When troubleshooting a fault code, use the source address displayed in INSITE™ electronic service tool to determine which ECM and circuit is affected.

Possible causes of this fault code include:

- Stage 2 fuel filter restriction high
- Pinched or damaged fuel supply line
- Stuck gerotor pump fuel pressure regulator located in high-pressure pump.

Refer to Troubleshooting Fault Code 4615. (</qs3/pubsys2/xml/en/procedures/122/122-t05-4615.html>)